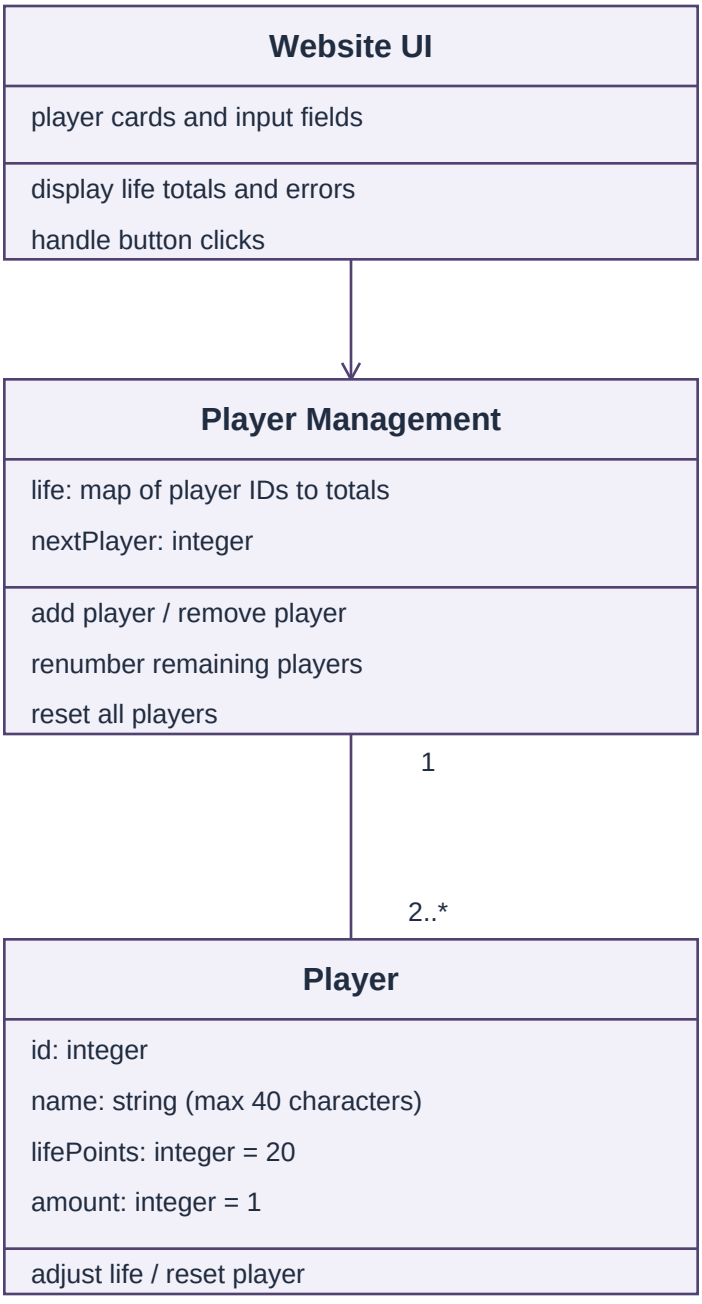
1. Class diagram

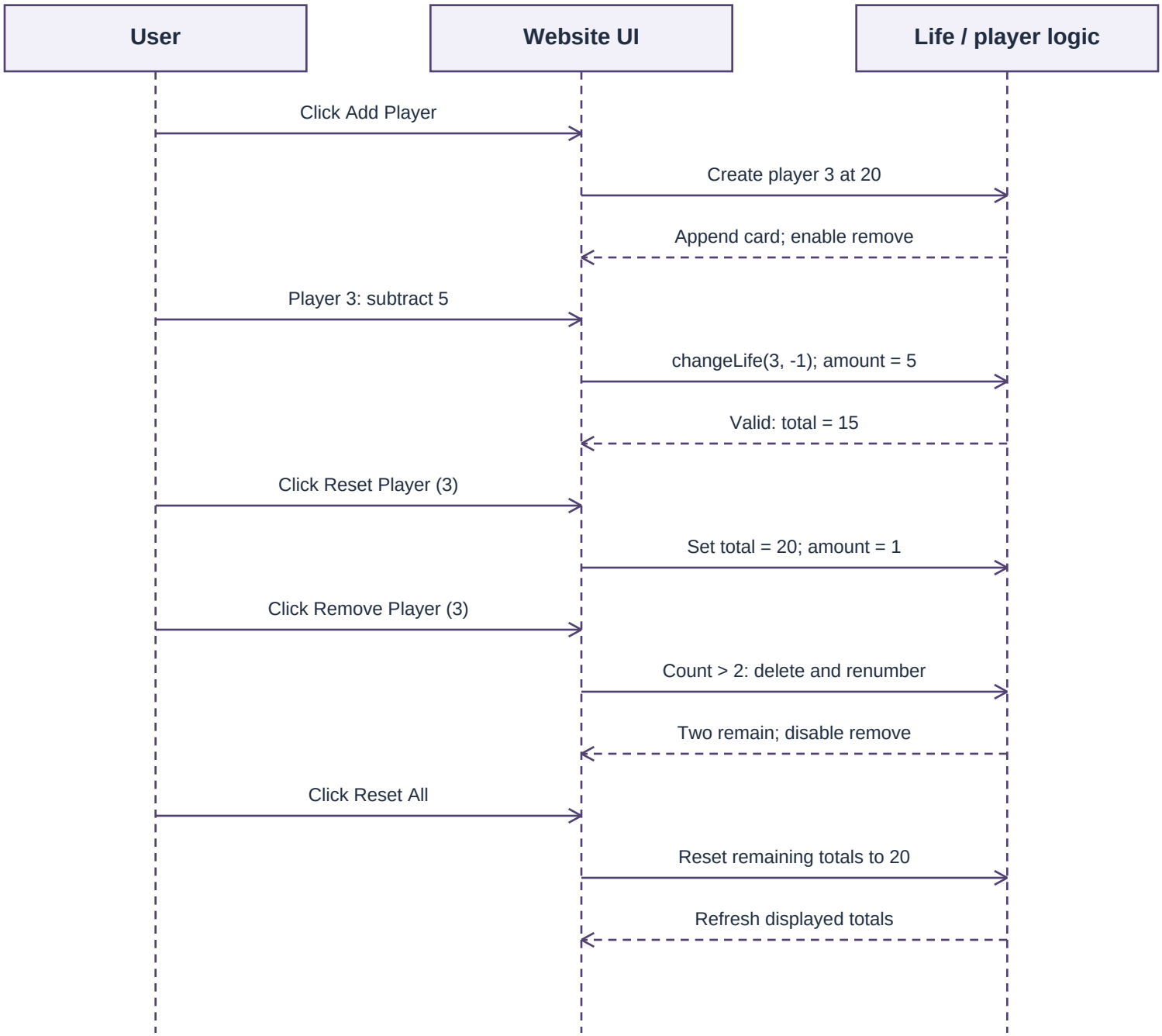
Conceptual design model - not literal JavaScript classes



The JavaScript stores life totals in the life object and names and amounts in HTML inputs. The conceptual Player groups these values; it is not an implemented class.

2. Sequence diagram

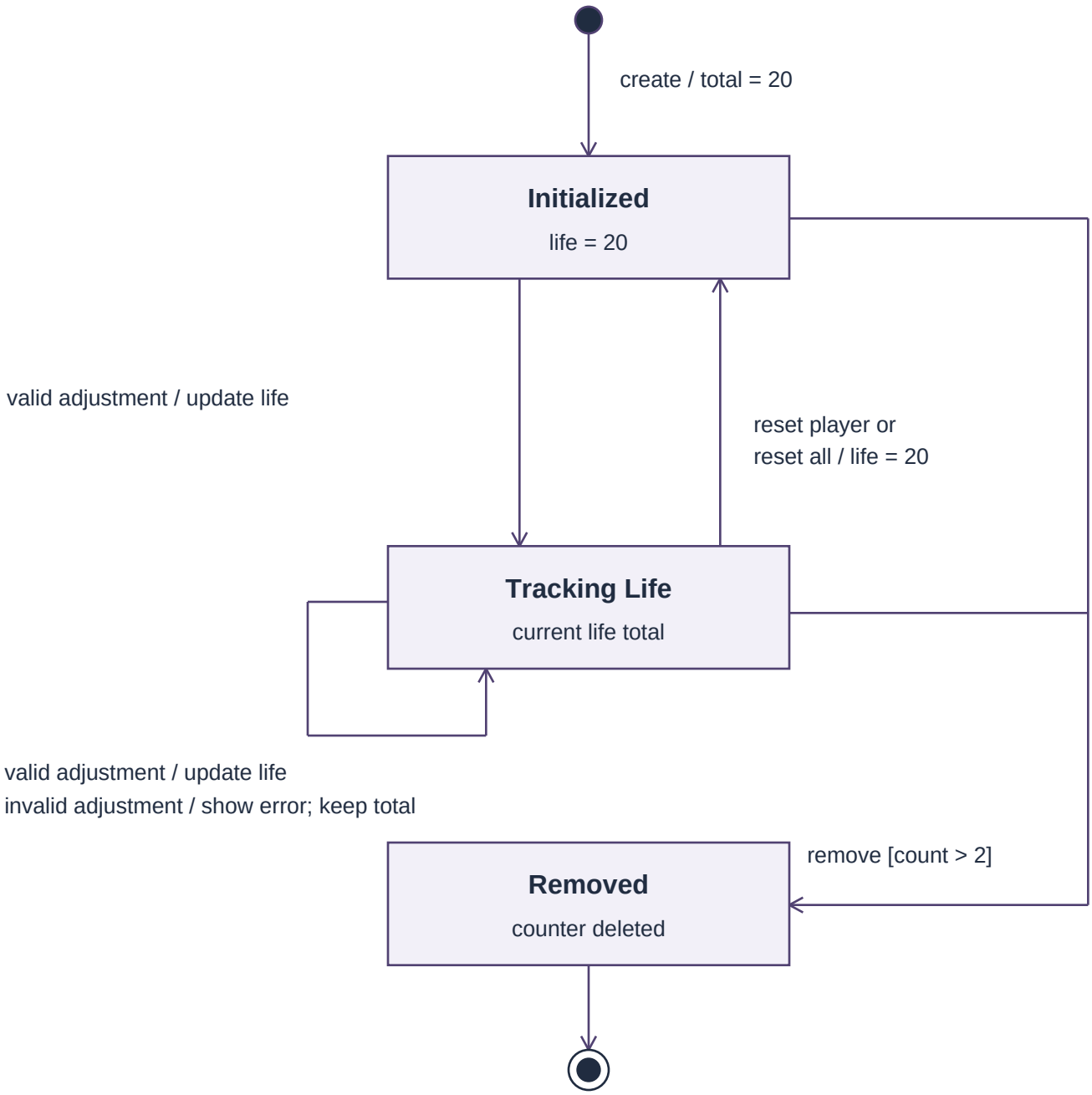
Scenario: add Player 3, subtract 5, reset, remove, then reset all



Invalid amounts or unsafe totals display an error without changing life. Reset actions preserve names and clear relevant errors. Removing a player is blocked when only two remain.

3. State machine diagram

Object modeled: one player counter in the current game



Reset while initialized leaves life at 20. An invalid adjustment leaves either active state unchanged. Removal is blocked at two players. Renumbering remaining players preserves their life totals and names.